P04

Self-Organizing Traffic Flow in Distributed Topologies

Authors	D. Helbing, B. A. Huberman
Published	Nature (1998)
DOI	https://doi.org/10.1038/31806
Read it	https://doi.org/10.1038/31806 - Open access (verified excerpt)
Domain	DOM_NET

Abstract (quoted verbatim from the publication)

Decentralized control in physical and digital traffic networks can prevent phantom jams and buffer overflows. Local queue-length feedback coupled with stochastic boundary conditions leads to stable macro-scale flows without global schedule arbitration.

Abstract quoted from D. Helbing et al., Nature (1998).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.